

CSC-355 Homework #2

substitutions, β -reductions, and recursion

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This homework is an opportunity to practice substitutions, β -converting terms, and recursion.

Problem 1: Substitution

Apply the substitutions to the terms.

- (a) $[w/y](xy)(yz)(zx)$
- (b) $[w/y]\lambda x.(\lambda y.xy)y$
- (c) $[w/y]\lambda x.(y\lambda y.xy)$
- (d) $[w/y]\lambda x.(\lambda y.xyy)$
- (e) $[w/y]\lambda x.y\lambda w.yxw$
- (f) $[w/y]\lambda w.(yww)$

Problem 2: β -reduction

Use β -reduction to rewrite each λ -expression into its normal form.

- (a) $\lambda x.x$
- (b) $(\lambda x.x)z$
- (c) $(\lambda x.x)(\lambda x.x)$
- (d) $\lambda x.\lambda y.(yx)(\lambda z.z)(\lambda w.(ww))$
- (e) $(\lambda x.\lambda y.x)(\lambda x.x)(\lambda x.\lambda y.\lambda z.(xy)(xz))$
- (f) Using our class' definition of "SUCC" and the church numeral for 3: SUCC 3

Problem 3: Recursion

Write a function in Python to compute 2^n , for some integer n . You must use the method of recursion we discussed in class. Here is the recursive definition of 2^n you should use:

$$2^n = \begin{cases} 1 & \text{if } n = 0 \\ 2^{n-1} \times 2 & \text{if } n > 0 \end{cases}$$